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Expt = 4(a) ¶

Title = All Pairs Shortest Paths Problem

In [3]:

```
def floyd(n, graph):
    for k in range(n):
        for i in range(n):
            for j in range(n):
                graph[i][j] = min(graph[i][j], graph[i][k] + graph[k][j])
    return graph

n = int(input("Enter the number of vertices:"))
print("Enter the adjacency matrix (enter 'inf' for infinity or '0' for no ed

graph = []
for i in range(n):
    row = []
    for x in input().split():
        if x.lower() == 'inf' or (x == '0' and len(row) != i):
            row.append(float('inf'))
        else:
            row.append(int(x))
    graph.append(row)

shortest_paths = floyd(n, graph)

print("==================================================")
print("Shortest paths between all pairs of vertices:")
for row in shortest_paths:
    print(*[x if x != float('inf') else 'inf' for x in row])
print("======================================================")
```

```
Enter the number of vertices:4
Enter the adjacency matrix (enter 'inf' for infinity or '0' for no edge):
0 3 inf 7
8 0 2 inf
5 inf 0 1
2 inf inf 0
==================================================
Shortest paths between all pairs of vertices:
0 3 5 6
5 0 2 3
3 6 0 1
2 5 7 0
======================================================
```

In [ ]:
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1
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1
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